

Why did Microsoft remove BinaryFormatter

Microsoft BinaryFormatter - Author not stated, Zenn.

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Introduction

From .NET 9 onwards, the serializer called BinaryFormatter has been removed completely. Engineers living today may not be familiar with it, but in the era of things like WCF (Windows Communication Foundation) it was used heavily, with rules such as putting [Serializable] on classes deriving from Exception. Even now you can see traces of it in the auto-generated code of WPF. Let us briefly look at how it is used. Preparation is complete simply by attaching [Serializable] to the target of serialization.

```

//just attach this attribute!
[Serializable]
internal class SampleData
{
    private readonly string name;
    private readonly int age;

    public SampleData(string name, int age)
    {
        this.name = name;
        this.age = age;
    }

    public override string ToString()
    {
        return $"name: {name}, age: {age}";
    }
}

```

Because it only has private fields, a modern serializer would require some attribute to be attached, but BinaryFormatter needs nothing at all. It can be restored easily with the code below.

```

var formatter = new BinaryFormatter();
var sampleData = new SampleData("Bob", 20);

using var stream = new MemoryStream();
formatter.Serialize(stream, sampleData);
stream.Position = 0;
//both name and age are restored
var readData = formatter.Deserialize(stream);
Console.WriteLine(readData.ToString());

```

It looks very convenient, but it ended up being removed on the grounds that it carries security risks. I will explain what kind of risks there are and why those risks came about, while looking at actual attack methods.

What kind of risks are there

According to Microsoft's documentation, using BinaryFormatter carries "vulnerabilities caused by deserializing untrusted data". This vulnerability is known as CWE-502.

The representative example given is that an attacker sends data over the network and, by deserializing it with BinaryFormatter, malicious code is executed.

As for whether it is safe if it is only used to save a settings file in the user's local environment, with no network involved, as stated in the following passage (excerpted from [The risks of assuming data to be trustworthy](#), bold by the author), it is better to think that **it can be used as a stepping stone, so it is no longer acceptable in the modern day.**

Consider also an app that uses BinaryFormatter to persist save state. This might at first seem to be a safe scenario, as reading and writing data on your own hard drive represents a minor threat. **However, sharing documents across email or the internet is common, and most end users wouldn't perceive opening these downloaded files as risky behavior.**

Why is there a risk

I said earlier that malicious code can be executed (RCE), but let us look at what factors actually make it possible. Note that there are vulnerabilities other than RCE, so various other attacks are also possible. Please be aware that dealing with this point does not mean it is fine to use BinaryFormatter.

Factor 1: the binary format of BinaryFormatter

Factor 1 lies in the format of BinaryFormatter. To get a feel for it, let us turn the `byte[]` after serialization into a string.

```
var formatter = new BinaryFormatter();
var sampleData = new SampleData("Bob", 20);

using var stream = new MemoryStream();
formatter.Serialize(stream, sampleData);
//forcibly output the data after serialization as ASCII
var payload = Encoding.ASCII.GetString(stream.ToArray());
Console.WriteLine(payload);
```

Then, although some noise is included, you can see that it contains "the assembly name plus the type name including SampleData", as below.

```
BinaryFormatterSamples, Version=1.0.0.0, Culture=neutral,
PublicKeyToken=null!BinaryFormatterSamples.SampleData
```

As for why it has this shape, BinaryFormatter writes and reads data according to the binary format called MS-NRBF below.

Of these, the "BinaryLibrary" is made to contain complete type information, and as far as I have investigated, ordinary serialization writes the data in the form of BinaryLibrary (complete type information) plus the actual data.

When reading as well, if the data was written with a BinaryLibrary, it tries to restore it according to the type information contained in it. This property, that **the type to restore can be freely specified from the data side** `[[1]]()`, is factor 1 that makes the attack possible.

Factor 2: the existence of abusable standard library types (Gadgets)

Among the standard libraries of C# there are several types that can be made to behave differently from their original intent when an instance is created. Malicious data using such types is called a Gadget. I will explain the outline using as an example

TypeConfuseDelegate, which uses `SortedSet<T>`, one of the representative Gadgets.

SortedSet has a `Comparer<T>` for comparing elements. This Comparer is called every time an element is added.

```
//constructor
public SortedSet(Comparer<T> comparer)
```

Let us try adding `"cmd.exe"` and `"/c calc.exe"` to `SortedSet<string>`.

```
var set = new SortedSet<string>();
set.Add("cmd.exe");
//comparer.Compare("cmd.exe", "/c calc.exe") is called for the comparison
set.Add("/c calc.exe");
```

I think you can vaguely see it now, but if you swap the implementation of comparer with another method that takes `(string, string)` and can do something bad, for example `Process.Start()`, then at the point of the second `Add` the calculator starts.

In other words, if you make full use of things like reflection `[[2]]()` to create a "SortedSet<string> whose comparer is really `Process.Start`, whose first element is "cmd.exe" and whose second is `"/c calc.exe"`" and save it with `BinaryFormatter`, then if you carelessly load it with `BinaryFormatter` it will obediently go and restore it and the calculator starts.

If you are interested, please look at the implementation below. I am impressed that anyone could come up with such wicked code.

The following is the most detailed on the principle. Types like this exist besides `SortedSet`, and because they are included in the standard library, countermeasures are difficult.

Countermeasures

Use a serializer that is considered safe (such as `System.Text.Json`). These serializers specify the type you want to restore at load time, and take the type information and constructor needed for restoration from it. The image is as below.

```
//takes the type information needed for restoration from T or the attributes attached to  
T, not from the stream  
var readData = serializer.Deserialize<T>(stream);
```

If files are already left in the user's hands, migration is not easy. In that case I think you will end up using `NRBFDecoder`, but honestly maintaining backward compatibility is not easy. It would be nice if there were a bit more documentation...

Summary

The fact that `BinaryFormatter` is not safe is touched on in quite a lot of articles, but there is not much explanation of why it is not safe, so I thought there might be demand for digging into that a bit deeper and wrote this.

I do not think `BinaryFormatter` is used much in modern products, but there is no guarantee that a vulnerability will not be found in the serializer you are using.

I think all we can do is live on while recognizing that the attacker's knowledge is far above ours, and being aware that the selection of a serializer includes security, not just compatibility and performance.

Footnotes

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There is also a format called `BinaryMethodCall`, and I suspect that saving with it would let you call any processing you like more directly, but I lack the ability to actually construct a payload. I await additional information from knowledgeable people. [↩]()

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It is not something that can be done easily. It looks like magic. [↩]()